# Chokepoints and Patience: Export Controls as Grand Strategy in the Semiconductor Contest

Viktor Draxler

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## Abstract

Export controls on advanced semiconductors and the tools to make them are often discussed as trade policy. This paper treats them as grand strategy: a bet that a lead in a few chokepoint technologies can be converted into time. We set out the conditions under which such a bet pays off (the chokepoint is real, allies hold the line, and the time bought is used) and argue that the third condition receives the least attention.

# 1 Argument

A chokepoint strategy has three parts: deny a rival a technology it cannot quickly make itself, keep partners aligned so the denial holds, and use the time bought to extend one's own lead. Most debate focuses on the first two. This paper argues that the third is the one that decides the outcome.

# 2 Conditions

| Condition | Question | Failure mode |
|---|---|---|
| Real chokepoint | Can the rival substitute within the planning horizon? | controls accelerate domestic substitution |
| Coalition | Do the few suppliers of the controlled items comply? | leakage through third countries |
| Use of time | Is the lead reinvested? | complacency behind the wall |

# 3 Time as the currency

Controls buy years, not permanence. Historical technology-denial regimes held for a decade or so before substitution or diffusion eroded them. The strategic question is what is built in that decade: manufacturing capacity, talent pipelines, and the next generation of tools.

# 4 Patience

Grand strategies fail when they demand quick results. Controls impose costs on the controlling side's firms immediately and deliver benefits slowly. A strategy that cannot survive two electoral cycles is not a strategy.

# 5 Historical analogies and their limits

Cold War technology-denial regimes are the obvious comparison. They held longest where the controlled items were embedded in complex supply chains that a single country could not rebuild, and they eroded fastest where the knowledge was in people rather than in machines. Semiconductor manufacturing sits at both extremes at once: tools are concentrated in a handful of firms, but process know-how moves with engineers.

| Era | Controlled domain | Approximate effective duration | What eroded it |
|---|---|---|---|
| Early Cold War | high-performance computing | 10-15 years | commercial diffusion of components |

| | | | |
|---|---|---|---|
| 1980s | machine tools and precision milling | about 8 years | third-country suppliers |
| 2020s | leading-edge chips and lithography tools | open question | substitution and stockpiling |

The table is illustrative; durations depend on how one defines "effective". The pattern is that denial buys time measured in years, and that diffusion through commercial channels is the usual end.

# 6 Coalition maintenance

Controls held by one country invite substitution by its partners' firms. The coalition question is therefore about compensation: who bears the lost revenue, and what do they receive in exchange? Arrangements that leave allied firms bearing costs with no share of the strategic benefit are unstable. The paper proposes that coalition partners be offered priority access to the capacity built during the window, which ties the third condition (use of time) to the second (coalition).

# 7 Indicators to watch

• Domestic tool substitution: share of a rival's new fabrication capacity using domestically produced tools.

• Talent flows: net movement of process engineers between the two ecosystems.

• Reinvestment: capital expenditure on next-generation capacity inside the coalition, relative to the pre-control baseline.

• Leakage: re-export volumes through third countries.

# 8 Conclusion

Measure the policy by what the time is used for, not by how long the wall stands.

# References

[1] Working paper; views are the author's own.